

tiplexing across channels, there is some type of charge build-up from channel to channel. You have to allow it to settle so that it does not affect your measurement accuracy. But with a dedicated ADC for every channel, we are creating systems with higher accuracy."

Modular and PC-based form-factors

"A multimeter with flexibility of being modular gives an upper hand as it can be clubbed with other equipment to enhance its usability in bigger applications," says Waghchoure. Especially for diagnostics, design and high-end applications, multimeters are always clubbed with other equipment.

Today, multimeters are available in a variety of form-factors like benchtop, handheld, USB modular, PXI modular and PC-based to fit in your application and testing requirements.

Siddique says, "For field usage handheld models that are easy to carry, portable and very user-friendly are recommended, while for automated environments there are USB- or PXI-based modular multimeters."

Thakkar adds, "For design applications or high degree of accuracy, a bench-top model can be chosen with closed eyes."

Depending on the application, sometimes you may require a multimeter in a form factor that can fit in racks, especially in automated test environments. Siddique says, "Customers may not look for big displays or panel at that time as they don't want to manually control and look at the instrument itself. Instead, they integrate various instruments and operate it from a single node like a computer. All the modular instrumentation today offers the equivalent of bench-top model specifications.

Green says, "Manufacturers have added USB and Ethernet interfaces to DMMs. Some have added



Tektronix benchtop DMM4050

the LXI protocol and incorporated Web browsers into the DMM to include data logging software and control of the DMM through a PC browser on the DMM's webpage."

Another form-factor gaining importance today is PC-based multimeters. Aggarwal says, "PC-based technologies don't just control the instruments, but the PC itself becomes the instrument."

In PC-based multimeters, there is a piece of hardware that can handle a lot of calculations and additional signal processing is done by the software. These are extremely easy-to-use, flexible systems bringing the advantage of bigger displays with more parameters and graphs, and data logging for further analysis.

Display technologies

Display is a key part of any T&M instrument. It is the first parameter or specification that appeals to the user and makes the instrument more user-friendly.

"Generally, a digital multimeter has a large and intuitive display that is simple to use and easy to read," says Sai Venkat, Pacific Marketing, Tektronix.

The earlier 8-LED displays have now been replaced with LCD and organic LED (OLED) displays in many devices. Siddique shares, "DMMs with unique OLED display provide 160-degree viewing angle and 2000:1 contrast ratio. That means you can read razor-sharp readings in pitch-dark and off-angle situations."

"DMMs show more in-

formation (graphics or statistics on measurements) on their displays. Reading rate has also increased substantially over the years," cites Green.

Communication interfaces

Connectivity is a key specification of every equipment today. There is a need to transfer data to store and analyse as well as view on a bigger screen. This has made wired and wireless connectivity an important component of the multimeter circuitry.

USB and Ethernet connectivity allows direct transfer of readings of various parameters. In PC-based equipment, all the features of a computer, like Internet connectivity, can be effectively used to share test and measurement results.

Innovative concepts like detachable displays or remote displays have gained popularity in the industry. A wireless, removable magnetic display allows you to be 9 metres (30 feet) away from the measurement point and is perfect for those difficult measurements where viewing the display is challenging. It gives users ultimate flexibility in unusual test situations. There is no more juggling of leads and the meter while stretching into tight spots. This technology uses a 2.4GHz ISM band wireless transmitter to immediately send measurements to the detached display, improving safety and increasing productivity for today's busy industrial and commercial electricians and electronic technicians.

Improved electronics circuitry

Refinement of the design using new technology tools has increased the performance of T&M equipment. With better circuitry and electronics, a lot of features can be integrated now. So while earlier multimeters could do only voltage and current measurements, new DMMs have far more capabilities.

Joshi cites, "Better circuitry and



Agilent U1253B True RMS OLED multimeter